



MECH MAINT- OFF SITE Department
SIMHADRI

Ref:SMTTP/MECH MAINT-/2022-23/LMIRESRV/171353

Date:

10/06/2022(07:43:14)

Subject:Review of LMI on “ Operation & Maintenance of reservoir at NTPC Simhadri”

Sir,

The Location Management Instruction (LMI) on " OPERATION AND MAINTENANCE OF RESERVOIRS" LMI/OGN/OPS/SYST/025 is done as per schedule.

This LMI is revised as per changes in department names & designations, same recorded in revision index

Further LMI is to be reviewed by EMD & operation for the changes in LMI in view of Floating Solar Power Generation Plant came in to service . Necessary changes may be incorporated in LMI .

Submitted for revision & Necessary changes in LMI by EMD & Operation .

Submitted by:

डी.श्री रमा सास्त्री - उप महाप्रबंधक

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On:10/06/2022(07:43:14) **Ref:**SMTTP/MECH MAINT-/2022-23/LMIRESRV/171353

No change seems to be required w.r.t non receipt of any intimation from Operation and EMD vide e-mail dated 10/06/2022.

Proposed LMI may please be approved.

Submitted by:

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On:23/06/2022(10:35:48) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMIRESRV/171353

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On:27/06/2022(12:50:23) **Ref:**SMTTP/MECH MAINT-/2022-23/LMIRESRV/171353

review and comment, if applicable

Submitted by:

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On:27/06/2022(18:16:50) **Ref:**SMTTP/MECH MAINT-/2022-23/LMIRESRV/171353

Following points may please be included in the LMI

1. The raw water reservoir level to be maintained between 2.6m to 3.0m to avoid damage to the Floating solar supporting ropes. --Responsibility: OPN
2. Grass cutting on bund to be done regularly to avoid cable fire hazard--Responsibility: CIVIL
3. Proper dressing of all cables to be ensured on Reservoir bund to avoid electrical shock and cable fire hazard--Responsibility: EMD
4. Water logging on north side of the reservoir to be avoided, proper draining and drain cleaning to be ensured. Responsibility: CIVIL

Submitted by:

मेडा नाग राजा - उप महाप्रबंधक

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On:28/06/2022(10:19:04) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMIRESRV/171353

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On:28/06/2022(19:53:38) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMIRESRV/171353

Forwarded for further approval

Submitted by:

राजशेखर पाला - अपर महाप्रबंधक

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On:01/07/2022(19:13:44) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMIRESRV/171353

Submitted for approval of competent authority

Submitted by:

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On:02/07/2022(10:40:48) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMIRESRV/171353

Please review the case file for LMI revision in light of the comments of EMD on page 4 of the above notesheet.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:15/07/2022(18:59:18) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMIRESRV/171353

EMD comments may please be reviewed for inclusion in the LMI.

Submitted by:

ASIS KUMAR MUKHOPADHYAY - 004586 (ADDL. GENERAL MANAGER ,MECH MAINT- OFF SITE)

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On:18/07/2022(12:46:30) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMIRESRV/171353

Revised & reviewed the points mentioned on page -4 of Note sheet & incorporated in LMI as per EMD comments.

Submitted by:

डी.श्री रमा सास्त्री - उप महाप्रबंधक

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On:20/07/2022(12:49:46) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMIRESRV/171353

Revised LMI may please be approved.

Submitted by:

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On:20/07/2022(18:43:02) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMIRESRV/171353

Please review the LMI from EMD side. In case of any clarification please contact DGM(OS). Once cleared from EMD end please send the file to undersigned for further approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:21/07/2022(08:47:42) **Ref:**SMTTP/MECH MAINT-/2022-23/LMIRESRV/171353

Reviewed

Submitted by:

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On:21/07/2022(15:27:30) **Ref:**SMTTP/MECH MAINT-/2022-23/LMIRESRV/171353

Reviewed and forwarded for approval please.

Submitted by:

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On:21/07/2022(16:52:10) **Ref:**SMTTP/MECH MAINT-/2022-23/LMIRESRV/171353

Submitted by:

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On:21/07/2022(18:59:40) **Ref:**SMTTP/MECH MAINT-/2022-23/LMIRESRV/171353

Approved & Submitted by:

गिरीश चन्द्र चौकसे - महाप्रबंधक

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On:22/07/2022(10:45:41) **Ref:**SMTTP/MECH MAINT-/2022-23/LMIRESRV/171353

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/SYST/025

ISSUE NO.1; REVISION NO. 2.00; DATE: JUNE 2022

OPERATION AND MAINTENANCE OF RESERVOIRS

APPROVED FOR IMPLEMENTATION

HOP(SIMHADRI)

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	<u>OPERATION AND MAINTENANCE OF RESERVOIRS</u>
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/SYST/025</u>
<u>2.b) REVISION/ISSUE NO.&DATE</u>	<u>ISSUE-1; REV.-2.00; JUNE; 2022</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>OGN/OPS/SYST/025/REV 0,MARCH 2020</u>
<u>4.PREPARED BY</u>	NAME: D Sri Rama Sastry DGM (MM/OS) SIGNATURE: -----
<u>5. CHECKED BY</u>	NAME: Asis Kumar Mukhopadhyay AGM (MM-OS/ Civil) SIGNATURE-----
<u>5.a) CHECKED BY</u>	Name : Piyaush Srivastava AGM (Chemistry) SIGNATURE:-
<u>5.b) CHECKED BY</u>	Name : Pala Rajasekhar AGM (Operation) SIGNATURE:-
<u>5.c). CHECKED BY</u>	Name : Dharmenndra Kumar Tomar AGM(EMD) SIGNATURE:_____
<u>7.RECHECKED BY</u>	NAME: Mathew E Kavoor GM - Mntce SIGNATURE-----
<u>7. APPROVED BY</u>	Rajesh Kumar Jha GM(O&M) SIGNATURE:-----
<u>8.DATE OF ISSUE</u>	June 2022
<u>9.INDEX LIST</u>	ENCLOSED

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1.0 INTRODUCTION

Simhadri Power Plant is provided with two nos reservoirs (One live reservoir and one dead reservoir) for storage of raw water necessary for the plant in order to provide a reliable and controllable water source for sustainable power generation. This raw water is fed to the reservoir from the yeleru canal through 4 nos Sweet water pumps located at Thadi village with 18 KM Raw Water pipe line.

The capacity of live reservoir is 50,000 cum and the capacity of dead reservoir is 7,63,000 cum. The raw water from reservoirs is supplied to the PT plant and the fire water storage tanks through raw water pipelines. Water quality changes with time in reservoirs depending on the storage and use conditions. Changes in operational practices of a reservoir can also influence the water quality. The reservoir, therefore, serves as a life line to the power station. Water reservoirs have been made to ensure supply of required quantity & quality of make up water during canal shut down. These canals are maintained by Govt Authorities, on which NTPC does not have any control. Thus, it becomes very important to keep the reservoirs in good working condition without any reduction in the designed capacity. This leads to Algae formation, accumulation of floating vegetation, hyacinth, dry leaves, deterioration in embankment etc, thereby affecting the reservoir capacity and affecting the water quality. This calls for regular and timely cleaning & maintenance of the reservoirs.

Problems associated with reservoirs :

1. Vegetational growth
2. Eutrophication (It is an increase in nutrients such as nitrogen and phosphorus that increase algal growth).
3. Falling and decaying of vegetation surrounding plant.
4. Contamination of reservoir water from nearby area.
5. Seepage/leakage
6. Damage of reservoir lining and embankment lining.

2.0 SCOPE

The scope of this LMI is to detail the various procedures in order to manage the reservoirs in an environmentally – sound manner, without deterioration in water quality with the objectives to:

1. To store & supply sufficient quality make up water through out the year.
2. To clean / remove the floating debris & silt.
3. To maintain the health of reservoir i.e. arresting seepage / leakage.
4. To Conserve precious water &
5. To reduce water charge & water Cess.

3.0 REFERENCE DOCUMENT

Operation guidance note OGN/OPS/SYST/025 Circulated by operation services _NTPC corporate center.

4.0 SUPERSEDED DOCUMENT

Issue -1, Rev-1

5.0 OPERATION AND MAINTENANCE PRACTICES TO BE FOLLOWED

Sl.No	Issues	Action	Frequency	Responsibility
1	Vegetation / water plant growth in reservoir	Regular cleaning and throwing away from the reservoir.	Yearly	MM OFS
2.	Growth of Trees / Bushes / Shrubs / Plants on the bund and in surrounding area.	Pruning / Trimming of Trees / Bushes / Shrubs / Plants of surrounding area & on bund to minimize falling of dry leaves etc into the reservoir. This will protect the embankment lining also.	Need based	MM/ CIVIL
3.	Using one reservoir for long period causing stagnation of water in the other one.	Both reservoirs are in parallel use to avoid siltation due to stagnating water.	Immediate	Operation
4.	Poor approach road condition + Area Lighting.	Motorable road should be ensured for inspection/ maintenance. Lighting Masts at suitable locations to be erected.	Joint report on quarterly basis.	MM/CIVIL/ OPERATION
5.	Human settlement	No human settlement should be permitted near reservoir in any case	Quarterly check	MM/CIVIL/ OPERATION
6.	Plant and township effluent should not be sent back to reservoir		Quarterly check	MM/ OPERATION
7.	Sampling of water quality	a)Physical and chemical measurements (PH,Conductivity,DO,Turbidity, Nitrate, Phosphates BOD, Ammonia	Weekly Quarterly	Chemistry

Sl.No	Issues	Action	Frequency	Responsibility
		b)Biological measurements (Coliforms)	Monthly	Chemistry
		c)Detailed water quality analysis including heavy metals and pesticides	Yearly.	
8	Arrangement for measurement of reservoir level	Suitable arrangement with marked levels should be provided at suitable location so that concerned people can take reading when ever required.	Immediate once provided levels to be painted yearly	MM
9	Seepage / Leakage of reservoir	Joint inspection of the bund	Quarterly	MM/CIVIL/OPERATION
10	Raw water reservoir level.	Raw water reservoir level to be maintained between 2.6m to 3.0 M to avoid damage to the Floating solar supporting ropes.	Regular	Operation
11	Grass cutting on bund.	Grass cutting on bund to be done regularly to avoid cable fire hazard	Monthly	Civil
12	Dressing of all cables to be ensured to avoid Electrical hazards	Proper dressing of all cables to be ensured on Reservoir bund to avoid electrical shock and cable fire hazard	Half yearly	Electrical
13	Water logging on sides of the reservoir to be avoided	Water logging on sides of the reservoir to be avoided, proper draining and drain cleaning to be ensure	Need based	Civil

6. EXCEPTION REPORTING:

The exceptions encountered during the inspection are to be reported as per the following format – MM/OPERATION/CHEMISTRY/O&M-CIVIL.

Description of exception	Area	Location of exception	Remarks

7.RESPONSIBILITIES

The responsibility of carrying out different tasks in relation to preventive measures, inspection and testing of above systems as detailed in the LMI are with HODs of MM/OPERATION/CHEMISTRY/O&M-CIVIL

8.RECORD KEEPING

The records to be maintained under this LMI include :

- a) Compliance record of the activities detailed under this LMI
- b) Exception reporting, if any.

The above records are to be maintained by HODs of MM/OPERATION/CHEMISTRY/O&M-CIVIL for the relevant responsibilities

9.0 REVIEW

This document shall be reviewed by HOP – Simhadri on Three yearly basis or as necessary

DISTRIBUTION LIST

Sl. No.	Date of issue	Issued to	Copy no.
1		GM(O&M)	
2		GM(Mntce)	
3		AGM (Opern)	
3		AGM (C&I-I/C)	
4		AGM(EM-I/C)	
5		AGM(OPN)	
6		AGM(O&M-CIVIL)	
7		SCE	
8		AGM(EEMG)I/C	

(for changes/revision)

01	02	02000622022	Change in department, designations, designations
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COMPLIANCE RECORD		
REF NO. LMI/OGN/OPS/SYST/025/		
SYSTEM / EQUIPMENT DESCRIPTION	ACTIVITY	COMPLIANCE